

MD. MONJURUL ISLAM

Software Engineer | Backend Focused

Uttara, Dhaka, Bangladesh | +880 1634072128 | Monjurulislamprottoy@gmail.com

[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Software Engineer with hands-on experience building web applications across the MERN and PERN stacks through personal and academic projects. Comfortable owning a feature end-to-end, from designing REST APIs and well-structured database schemas to implementing role-based access control and real-time functionality. Particularly drawn to the backend, writing server-side logic that stays clean and easy to reason about as an application grows.

TECHNICAL SKILLS

- **Frontend:** JavaScript (ES6+), React.js, Tailwind CSS, HTML5, CSS3
- **Backend:** Node.js, Express.js, WebSockets, Appwrite, Firebase
- **Database & ORMs:** MongoDB, PostgreSQL, Redis, Cloudinary, Mongoose, Prisma
- **State Management:** Redux Toolkit, React Hook Form, Multer
- **Deployment:** Docker, AWS EC2, AWS CloudFront, Vercel
- **Tools & Architecture:** Git, GitHub, Postman, RESTful APIs, Gemini API Integration

PROJECTS

HealthBridge – Healthcare Management System | *React, Redux Toolkit, Node.js, Express, MongoDB, Redis, Socket.io* [Live](#) | [GitHub](#)

- Built a role-based system for Patients, Doctors, Staff, and Admins using JWT auth and custom access control.
- Used Redis locks (SETNX) during slot booking to stop two users from booking the same appointment slot.
- Integrated Google Gemini API to analyze patient symptoms and decode prescription images, protected by Redis rate limiting.
- Implemented real-time doctor consultation queues and dynamic 48-hour follow-up chats using Socket.io and Redis.
- Managed complex frontend state using Redux Toolkit and Redux Persist, handling global auth sessions and optimistic UI updates.
- Deployed Dockerized backend on AWS EC2 with CloudFront for secure HTTPS routing.

GroFresh – Full-Stack E-commerce Platform | *React, Node.js, Express, MongoDB* [Live](#) | [GitHub](#)

- Built a full-stack e-commerce web application with separate panels for buyers and sellers.
- Built scalable Node.js/Express APIs to manage authentication, product listings, and cart workflows.
- Integrated Stripe payment gateway to process secure online transactions during checkout.
- Used Multer and Cloudinary for uploading and storing product images.

EDUCATION

Daffodil International University

Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering; CGPA: 3.06

-2026

- **Relevant Coursework:** Data Structures, Algorithms, Object-Oriented Programming, Database Management Systems, Web Engineering.

THESIS

Predictive Modeling of AI Adoption: A Hybrid Approach – An analytical study examining the behavioral drivers of advanced AI adoption among engineering students; bridges traditional statistical modeling with modern predictive algorithms to forecast technology acceptance.

ACHIEVEMENTS & EXTRACURRICULARS

- **Competitive Programming:** Participated in the Unlock the Algorithm Competition (2023) and Take-Off Problem Solving Competition (2022) at DIU.
- **Tech Communities:** Executive Member of the DIU Computer Programming Club (CPC) and Member of the DIU Robotics Club.